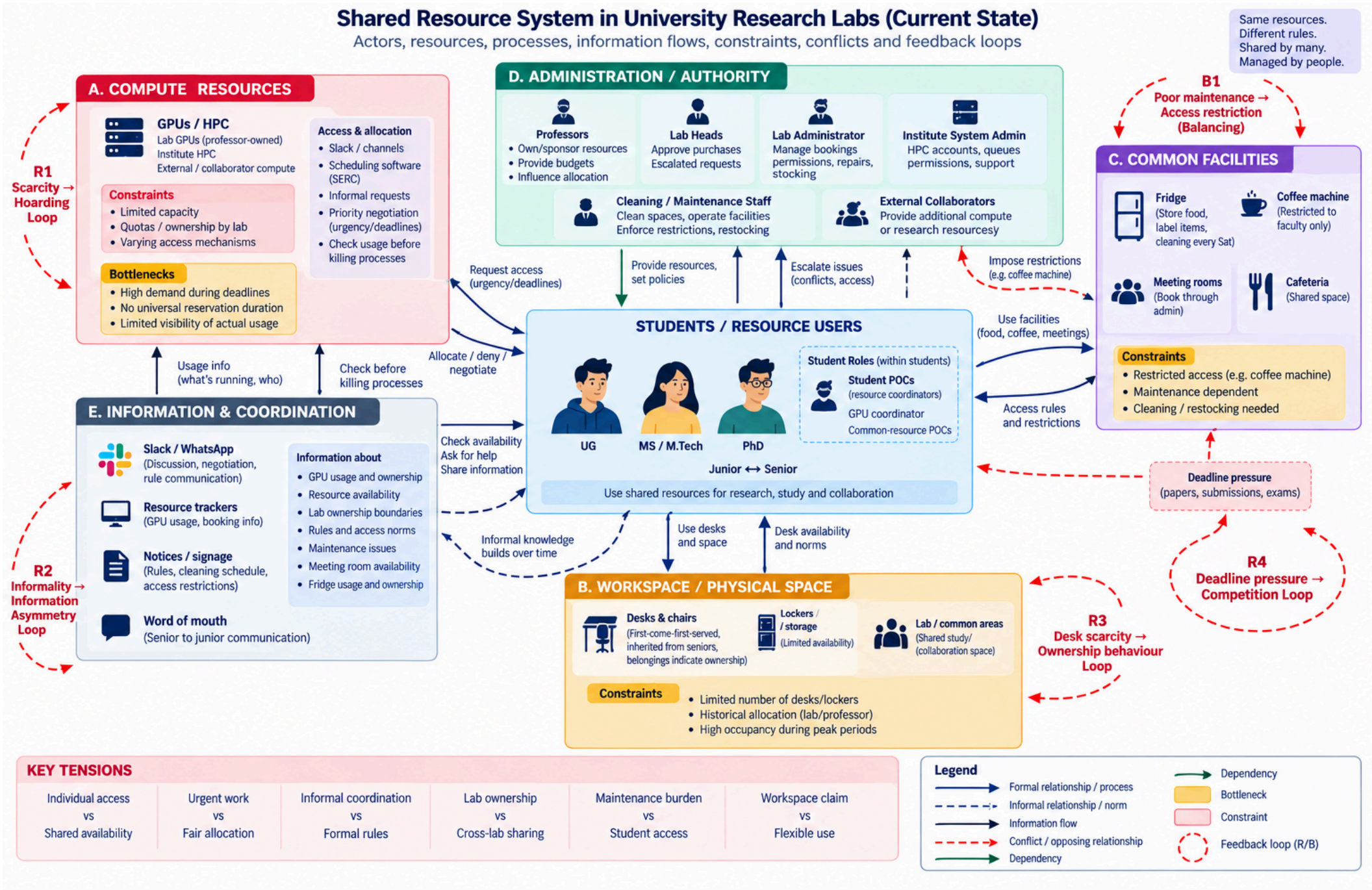
SYSTEMS THINKING & DESIGN - PROJECT REPORT

Systemic Analysis of Shared Lab Resource Governance

PHASE 2 SUBMISSION

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System Map


Shared Resource System in University Research Labs (Current State)
Actors, resources, processes, information flows, constraints, conflicts and feedback loops
Same resources.
Different rules.
Shared by many.
Managed by people.
A. COMPUTE RESOURCES
GPUs / HPC
Lab GPUs (professor-owned)
Institute HPC
External / collaborator compute
Access & allocation
Slack / channels
Scheduling software (SERC)
Informal requests
Priority negotiation (urgency/deadlines)
Check usage before killing processes
Constraints
Limited capacity
Quotas / ownership by lab
Varying access mechanisms
Bottlenecks
High demand during deadlines
No universal reservation duration
Limited visibility of actual usage
R1
Scarcity → Hoarding Loop
D. ADMINISTRATION / AUTHORITY
Professors
Own/sponsor resources
Provide budgets
Influence allocation
Lab Heads
Approve purchases
Escalated requests
Lab Administrator
Manage bookings permissions, repairs, stocking
Institute System Admin
HPC accounts, queues permissions, support
Cleaning / Maintenance Staff
Clean spaces, operate facilities
Enforce restrictions, restocking
External Collaborators
Provide additional compute or research resourcesy
B1
Poor maintenance → Access restriction (Balancing)
C. COMMON FACILITIES
Fridge
(Store food, label items, cleaning every Sat)
Coffee machine
(Restricted to faculty only)
Meeting rooms
(Book through admin)
Cafeteria
(Shared space)
Constraints
Restricted access (e.g. coffee machine)
Maintenance dependent
Cleaning / restocking needed
Request access (urgency/deadlines)
Provide resources, set policies
Escalate issues (conflicts, access)
Impose restrictions (e.g. coffee machine)
Use facilities (food, coffee, meetings)
Access rules and restrictions
Allocate / deny / negotiate
Usage info (what's running, who)
Check before killing processes
STUDENTS / RESOURCE USERS
UG
MS / M.Tech
PhD
Junior ↔ Senior
Student Roles (within students)
Student POCs
(resource coordinators)
GPU coordinator
Common-resource POCs
Use shared resources for research, study and collaboration
Deadline pressure
(papers, submissions, exams)
E. INFORMATION & COORDINATION
Slack / WhatsApp
(Discussion, negotiation, rule communication)
Resource trackers
(GPU usage, booking info)
Notices / signage
(Rules, cleaning schedule, access restrictions)
Word of mouth
(Senior to junior communication)
Information about
GPU usage and ownership
Resource availability
Lab ownership boundaries
Rules and access norms
Maintenance issues
Meeting room availability
Fridge usage and ownership
Check availability
Ask for help
Share information
Informal knowledge builds over time
R2
Informality → Information Asymmetry Loop
Use desks and space
Desk availability and norms
B. WORKSPACE / PHYSICAL SPACE
Desks & chairs
(First-come-first-served, inherited from seniors, belongings indicate ownership)
Lockers / / storage
(Limited availability)
Lab / common areas
(Shared study/ (collaboration space)
Constraints
Limited number of desks/lockers
Historical allocation (lab/professor)
High occupancy during peak periods
R3
Desk scarcity → Ownership behaviour Loop
R4
Deadline pressure → Competition Loop
KEY TENSIONS
Individual access vs Shared availability
Urgent work vs Fair allocation
Informal coordination vs Formal rules
Lab ownership vs Cross-lab sharing
Maintenance burden vs Student access
Workspace claim vs Flexible use
Legend
Formal relationship / process
Informal relationship / norm
Information flow
Conflict / opposing relationship
Dependency
Dependency
Bottleneck
Constraint
Feedback loop (R/B)

Formal and Informal Structure

Type	Structure
Formal–institute	4-vs-1 GPU quotas; 6-hour interactive cap; institute AWS; HPC/admin tickets; 2 meeting rooms via admin (with a possible website)
Formal–institute	Centralized AC tied to faculty presence; off nights/Sundays
Formal–lab	Faculty budgets and sponsorship
Formal rule (written)	Coffee machine: staff-operated, faculty/guest only, enforced by a guard; fridge "SERC" label; Saturday auto-disposal notice
Formal–service	Cleaning staff execute but never decide; they request stock from admin department; supervisor in building
Quasi-formal (emerging)	New desk name-writing system (recent, caused by crowding); faculty-named desks (new this year); SERC ownership website; single student GPU manager; PhD POCs
Informal	First-come-permanent desk claims; unwritten return-priority; courtesy of messaging before occupying during long absence; jacket-on-chair identification
Informal	Urgency priority by persuasion on Precog Slack ("you have to convince someone that I want it more"); ask-first before killing GPU runs
Informal	Desk inheritance from graduating seniors; junior → senior → faculty escalation ladder; WhatsApp as the complaint sink ("we'll sort it ourselves")
Informal boundary	"You can't come to this side and sit [firmly]" - cross-lab desk line maintained by common understanding and by anticipated professorial disapproval, not by any posted rule

Bottlenecks

Bottleneck	Where the flow gets stuck	Impact on the system
GPU scheduling during high demand	Multiple users need compute at the same time, especially near deadlines	Experiments wait, work gets delayed
Idle GPU reservations / resource hoarding	GPUs remain occupied even when not actively being used	Effective availability drops and urgent users get blocked
Unclear GPU ownership or usage visibility	Users may not immediately know who is running a process or whether it is still needed	Extra coordination time, risk of accidental interruption, conflict
Manual negotiation for priority	Users have to message or convince each other when multiple people need the same resource	Allocation takes time and depends on social coordination rather than a fast process
Desk allocation during peak occupancy	More people want to work in the lab than there are immediately usable desks	Some students cannot find a workspace
Temporarily empty but socially occupied desks	A desk may be physically free but still considered someone's desk	Available physical capacity is not fully utilized
Admin request processing	Access or technical requests depend on a small administrative process/team	Requests wait before users can continue work
Meeting-room booking clashes	Multiple groups need the same rooms at overlapping times	Meetings or evaluations have to move or wait
Common-space takeover during evaluations/events	Cafeteria/chairs/shared areas get temporarily occupied for academic activities	Normal users lose access to those spaces

Maintenance/restocking response time	Shared facilities need admin/staff intervention when something runs out or breaks	Resource remains partially or fully unusable until serviced
Conflict resolution through informal channels	Small disputes are handled through messages, seniors, or peer negotiation	Resolution speed varies depending on people involved
Peak-period congestion	Exams, project deadlines, and submissions cause many users to demand the same resources together	Existing queues become much worse

Tensions

Tension	Side A	Side B	Resulting behaviour
Fairness vs urgency	Everyone should get equal access	Deadline-critical work may deserve priority	GPU priority gets negotiated informally
Individual ownership vs shared utilization	“This is my desk/GPU slot”	Empty/idle resources should be used by others	Informal permission systems emerge
Reservation vs utilization	Reserve resources early to protect future work	Resources should only be held when actively needed	GPU hoarding and idle reservations
Lab ownership vs institute efficiency	Resources belong to the lab/professor who funded them	Sharing across labs could improve utilization	Strict boundaries around compute resources

Senior familiarity vs newcomer equality	Seniors understand and have historically used the space	New students also need fair access	Junior hesitation and informal hierarchy
Formal authority vs informal coordination	Admin/faculty could enforce rules	Students prefer resolving small issues themselves	Most conflicts stay informal
Flexibility vs clarity	Unwritten norms allow adaptation	Explicit rules reduce confusion	New students struggle to know what is allowed
Personal desk ownership vs space scarcity	Students want stable personal workspaces	There are more users than desks	Empty desks may remain psychologically “owned”
Convenience vs cleanliness	Students want to eat/work where convenient	Food and waste disturb shared workspace	Complaints about food left on desks
Access vs maintenance	Students want unrestricted access to shared facilities	Facilities need maintenance and responsible usage	Access can be restricted
Student access vs faculty priority	Students want coffee machine/ common-resource access	Facilities are sometimes reserved for faculty/staff	Access becomes unequal
Late-night work vs energy conservation	Students work late in labs	AC/electricity is switched off to reduce consumption	Poorer working conditions at night
Current demand vs historical allocation	Student population changes	Physical resources remain allocated according to older arrangements	Some labs become overcrowded while allocations remain fixed

Feedback Loops (Preliminary Observations)

- ☐ **Desk crowding → name-writing rule → less confusion.** The lab got crowded, tables became scarce, people started writing their names on desks for the first time, and occupancy became clearer.

- ☐ **Blind job-killing → ownership-check habit → fewer accidents.** People were risking killing each other's running GPU jobs, so they started checking the website and asking the runner before acting; that habit prevents the problem that created it.

- ☐ **Spoiled food → Saturday-clearing notice → empty fridge.** Food left too long became a hygiene issue; a notice was put up, and every Saturday the fridge is emptied. The problem keeps feeding the practice that removes it.

- ☐ **Bad coffee maintenance → access removed → maintenance problem vanishes.** Students failed the upkeep rota they had agreed to, so access was taken away, staff took over, and the original problem disappeared - along with the benefit.

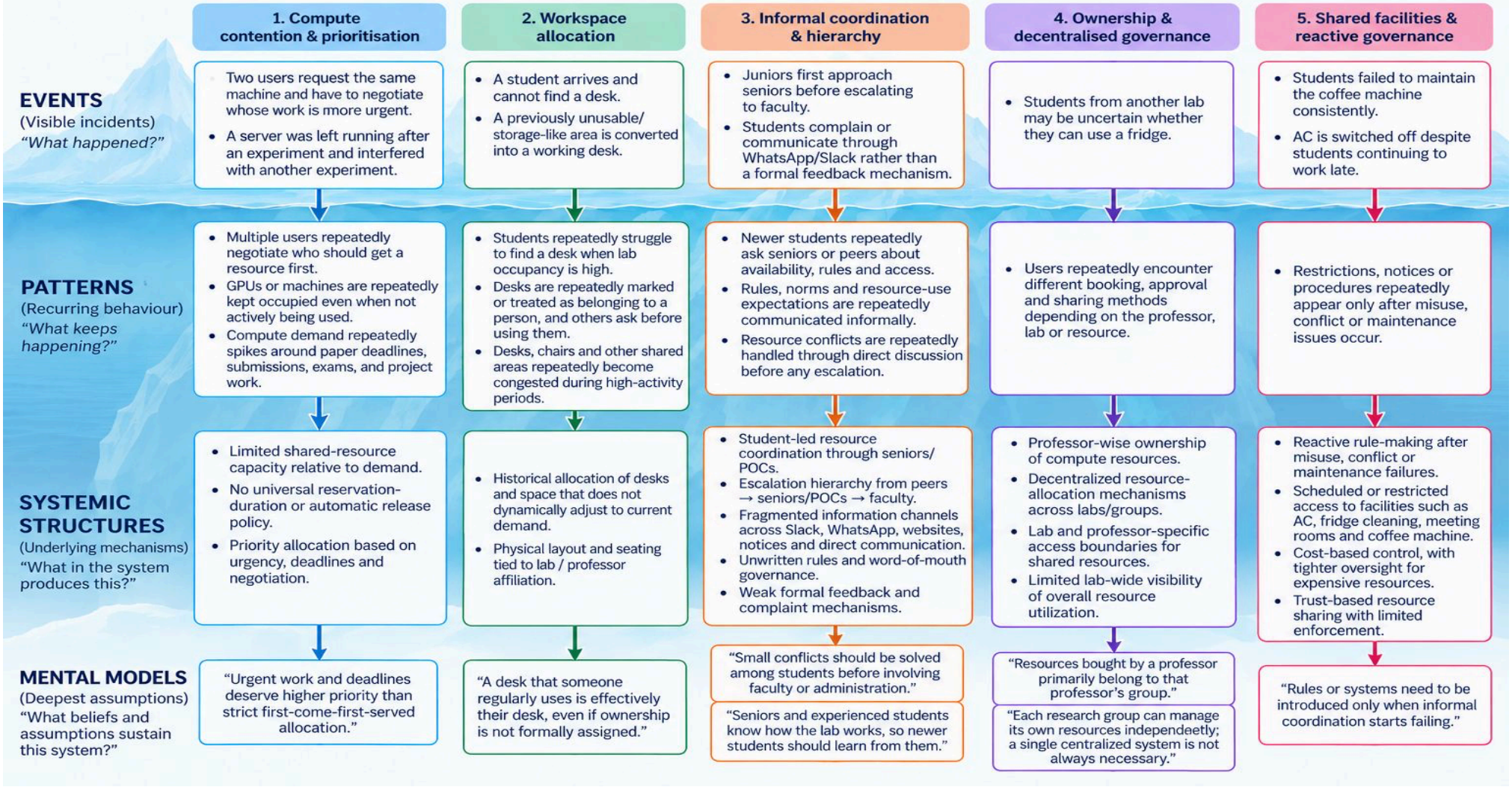
- ☐ **Urgent need → direct request → negotiated adjustment.** Someone who needs a GPU or a desk urgently asks the current holder directly; the holder adjusts; the urgency passes. This works as long as the holder is reachable and willing.

- ☐ **Deadlines and summer heat → crowding → stronger desk claims → tighter crowding.** Peak periods (exams, heat escape) fill the labs, contested space makes people defend "their" desk more strongly, and long-held claims make the next peak even tighter.

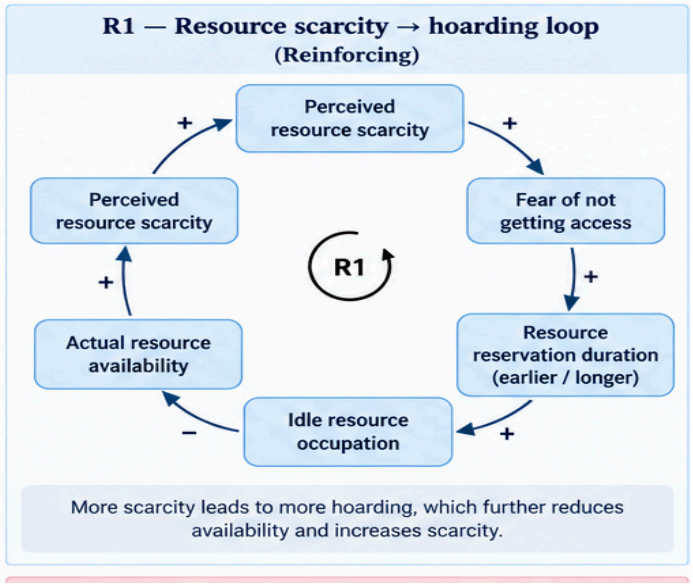
- ☐ **Visible damage → individual asks a senior faculty member → repair.** The light/fan-less desk got fixed only when someone personally raised it; same for junk removal. Here the trigger works through one person's initiative rather than any channel.

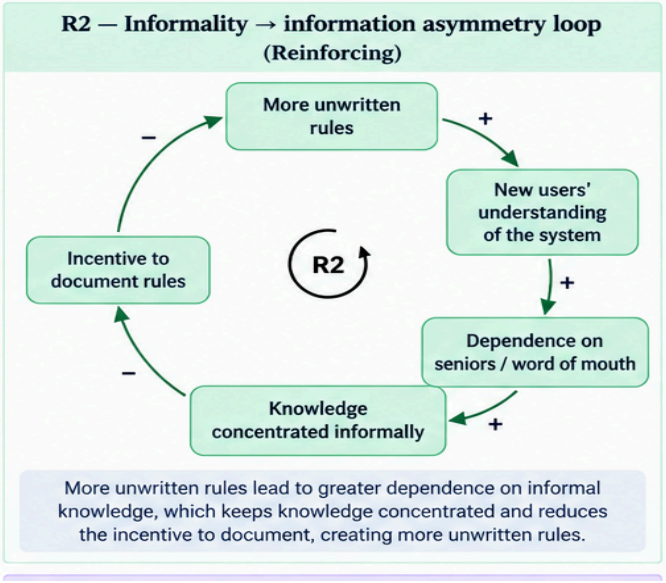
Iceberg Model of Lab Resource Use

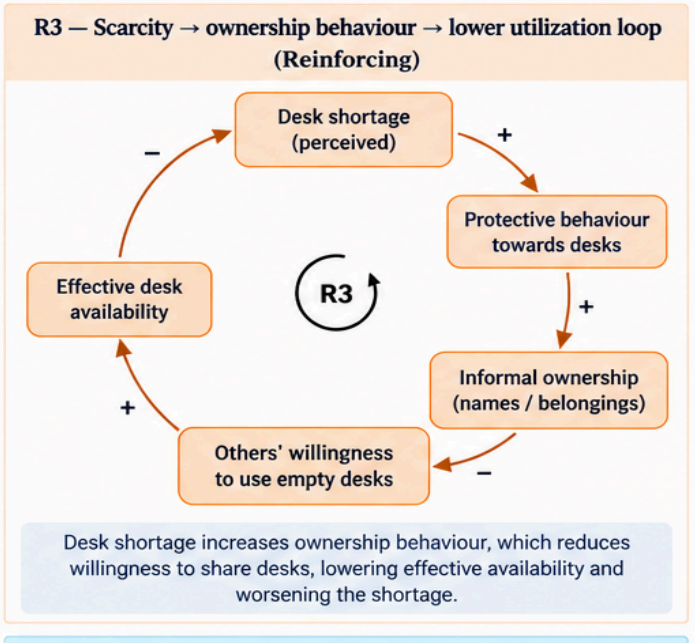
From visible incidents to underlying structures and mental models

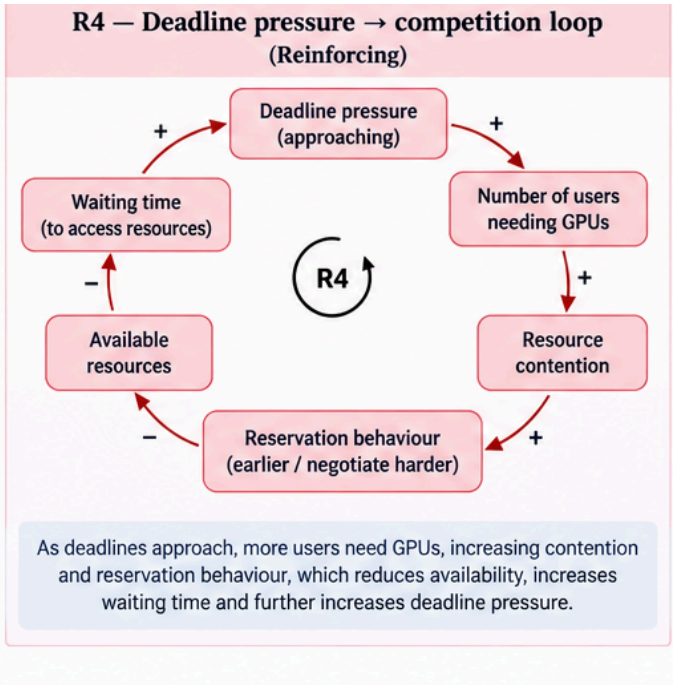


Causal Loops/ Feedback Loops

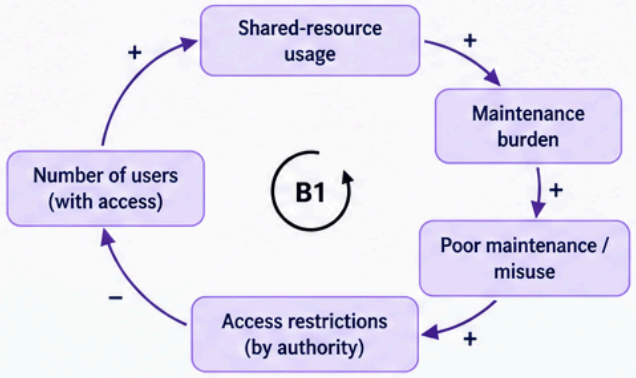








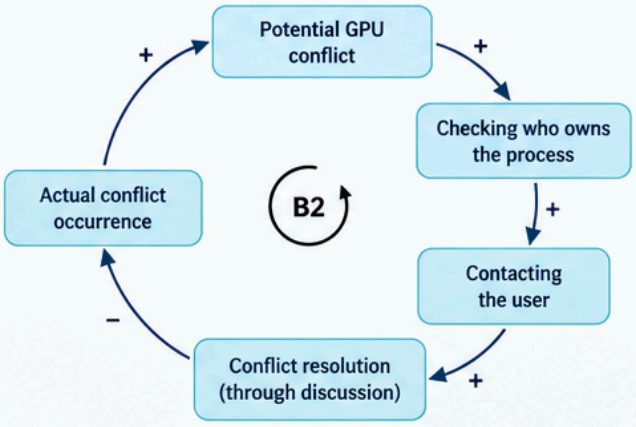
B1 — Poor maintenance → access restriction loop (Balancing)



Higher usage increases maintenance burden, leading to poorer maintenance and access restrictions, which reduce the number of users and bring usage (and maintenance burden) down.

Side effect (not a separate loop): Access restrictions → lower student access → perception of unfairness / under-utilization.

B2 — Communication → conflict avoidance loop (Balancing)



When a potential conflict arises, users communicate and resolve it, which reduces actual conflicts, lowering future potential conflicts.

Unintended Consequences

Unintended consequence	How it emerges
Artificial scarcity despite available capacity	Users reserve GPUs or desks defensively, so resources may be technically available but practically unavailable to others
Resource hoarding becomes rational behaviour	When future access is uncertain, users are incentivized to hold resources longer than necessary
Urgent work can become harder to prioritize fairly	Informal negotiation helps flexibility, but priority may depend on persuasion, relationships, or who speaks first
Junior/new users face an information disadvantage	Because many rules are unwritten, newer members depend on seniors to understand what they can use and how
Informal hierarchy develops without formal authority	Seniority, familiarity, and long-term presence can create perceived ownership or greater influence over shared resources
Physical resources become socially unavailable	A desk may be empty but still treated as someone's desk because of names, belongings, or historical use
Capacity appears lower than it actually is	Persistent desk claims and idle GPU reservations reduce effective utilization
Different groups receive unequal access experiences	Professor-wise ownership and decentralized systems mean access depends on which group a student belongs to
Cross-lab resource pooling is discouraged	Resources can be underused in one group while another group experiences scarcity because access boundaries remain separate
Relationships become a substitute for formal access	Knowing a senior, POC, or collaborator can make it easier to navigate or gain access to resources

Minor problems remain invisible to decision-makers	Students usually solve issues among themselves, so faculty/admin may never see recurring systemic problems
The system learns slowly	Complaints stay in Slack, WhatsApp, or conversations instead of becoming structured feedback that changes policy
Rules emerge only after failure	Misuse or conflict happens first, and only then are restrictions, notices, or guidelines introduced
Collective punishment can replace targeted correction	Misuse by some students can lead to access being removed for all students, as happened with the coffee machine
Successful informal coordination hides structural weaknesses	Because students often resolve conflicts peacefully, the underlying shortage or governance problem can appear less serious than it is
Manual coordination load shifts onto a few students	Senior students/POCs become informal resource managers, creating dependence on them and extra invisible work
POCs can become bottlenecks themselves	When approval or coordination depends on one or a few people, increasing resource demand can overwhelm them
Automation can become fragmented rather than systemic	One group creates a tracker while another continues using Slack or direct messages, producing multiple parallel systems
Local optimization can reduce system-wide efficiency	Each professor/group optimizes its own resources, while overall SERC utilization may remain inefficient
Urgency can be strategically overstated	If “urgent” requests receive higher priority, users may have an incentive to frame their work as urgent

FCFS can reward early reservation rather than genuine need	Users who know their future requirements earlier can block those who later develop more important needs
Trust-based systems can work well until misuse scales	As the number of users increases, trust alone becomes harder to sustain and stricter control may suddenly appear
Stricter control can reduce user ownership	Once staff or faculty take over a previously shared resource, students may feel less responsible for maintaining it
Resource restrictions can encourage workarounds	Users may borrow another person's access, use another lab's name, or rely on informal arrangements instead of formal access
Shared spaces gradually become personalized territory	Desks, chairs, and storage accumulate personal belongings, reducing their flexibility as shared resources
Temporary workarounds become permanent norms	Practices originally created to solve short-term scarcity-like writing names on desks-can become the default system
Growth increases governance complexity nonlinearly	A process that works with a small group can fail quickly when more users and machines are added
Peak-period congestion becomes self-reinforcing	As deadlines approach, users reserve earlier and longer, making availability even worse during the exact period of highest demand
Underutilized resources and unmet demand coexist	One group may have unused compute while another cannot access enough, because ownership and coordination are fragmented

Fairness becomes subjective	Without common rules, different users may interpret fairness as FCFS, equal access, seniority, urgency, ownership, or research importance

Root Causes

Priority	Category	Root cause	Why this is a root cause
1	**Structural**	**Fragmented ownership and governance of resources**	Resources are owned and controlled by different professors/groups, so allocation, access, and decision-making stay siloed. This drives many downstream issues: uneven access, duplicated processes, underutilization in one group, and scarcity in another.
2	**Structural**	**Dependence on informal social coordination**	The system relies heavily on word of mouth, seniority, courtesy, direct messaging, and personal negotiation. Because this mechanism governs access and conflict resolution, it repeatedly produces inconsistency, newcomer disadvantage, and hidden hierarchy.
3	**Structural**	**Lack of unified visibility and coordination**	There is no single reliable view of who is using what, what is available, how long it is reserved, or what the priority is. This directly creates avoidable waiting, confusion, idle occupation, and coordination overhead.
4	**Structural**	**Demand–capacity mismatch at peak times**	Critical resources such as GPUs, desks, chairs, and spaces cannot always absorb peak demand. This is a root driver of competition, reservation pressure, urgency conflicts, and workarounds.
5	**Systemic tension**	**Local autonomy vs system-wide efficiency**	Groups want control over resources they fund and manage, but the larger system benefits from pooling and sharing. This unresolved tension sustains fragmentation and prevents global optimization.

6	**Systemic tension**	**Flexibility vs standardization**	Informal, case-by-case decisions make the system adaptable, but the same flexibility prevents consistent rules and predictable access. This tension keeps the system from becoming fully standardized.
7	**Systemic tension**	**Fairness vs urgency**	Equal or FCFS access conflicts with prioritizing deadline-critical work. Because both are legitimate goals, priority decisions repeatedly become negotiable and subjective.
8	**Structural**	**Historical allocations do not adapt to changing demand**	Desk, space, and resource allocations continue even when group size and usage change. This creates a persistent mismatch between where capacity exists and where demand actually is.
9	**Structural**	**Weak feedback-to-policy conversion**	Problems are often solved locally in chats or conversations, so recurring issues do not consistently become institutional learning or policy change. As a result, the same issues reappear.
10	**Systemic tension**	**Trust vs control**	The system wants low-friction, trust-based sharing, but misuse and scarcity create pressure for monitoring and restrictions. This tension explains why access often becomes stricter only after problems occur.

Reframe the Problem

Challenging Initial Assumptions

The Phase 1 study framed the problem as *how resource boundaries, claim persistence, uneven information and distributed responsibility turn nominal capacity into unequal usable access*. Before reframing, we audited the beliefs embedded in that framing - and in our own early reading of the problem - against the interview evidence. Each assumption below is kept, modified, or discarded with one anchor quote.

Assumption	Verdict	Evidence
The core problem is not enough resources	**Modified** - capacity exists but is unreadable and non-releasing	GPUs held weeks in advance: "they just want to occupy it so that they might run the experiment some day"; prime desks vacant daily while claimed
There are no written rules	**Discarded** - rules exist but are fragmented, mostly unwritten, and appear only after breakdowns	Saturday fridge-clearing notice, coffee window 9–12, lab-labelled fridge; desk name-writing introduced only this year after crowding
Users are selfish; that is why resources degrade	**Discarded** - helpfulness is high, but offers of help never reach those who need them	"Everyone, 95%, will be willing to go out of their way to help… if someone requires genuine [need]"
Juniors can't use desks because seniors forbid it	**Modified** - access is usually granted when asked; the barrier is the belief asking will fail	"I still have this fear that I cannot [sit there]" - from a respondent never actually refused
The SERC–Precog boundary is the fault line	**Discarded** - boundaries differ per resource, so users hold multiple conflicting maps of what is shared	One floor, three fridge regimes; a "hard rule, no exceptions" on compute that follows the professor, not the lab

Formal authority doesn't care about student problems	**Discarded** - authority responds when reached; signals simply don't arrive	Broken light/fan/AC fixed immediately when one student approached faculty
Peaceful settlement means everyone's needs are met	**Modified** - peace is real but one-directional: juniors yield rather than negotiate	"We sort it among ourselves… why would I bother anyone about a very small situation?"
The system is stable; problems are recent congestion	**Modified** - the floor is mid-transition, formalising informal norms only when crowding makes failure visible	"Ye abhi recently hua hai… abhi hum log thoda conscious ho chuke hain"

Reading the pattern. The discarded assumptions all located the problem in *people or quantities* - selfishness, under-supply, indifferent authority. The surviving ones all locate it in *structure*: physical emptiness is not readable as availability, permission is informally real but never felt, expiry mechanisms exist only where decay is visible (food), and no channel returns an unserved need to anyone authorized to act on it. The system does not prevent access; it simply never informs the people who could enable it.

Comparing Stakeholder Perspectives

Each stakeholder group in this system describes the *same* resources through a different lens - the divergence itself is evidence for Phase 1's finding that information is unevenly distributed. The comparison below uses one anchor per group; quotes are verbatim from the interviews.

Resource / Issue	SERC students	Juniors & newcomers	Precog students	Cleaning staff
Desks	A settled norm - "no booking system, only name writing system"; courtesy is asking first	A site of anxiety - "I still have this fear" of senior tables, despite never being refused	A scarcity - 30–40 students, more people than desks; two juniors have none: "if everyone is sitting, there's no solution"	Not their domain; concerns are broken/dirty furniture, not claims
Fridge & amenities	Lab-labeled, with informal exceptions;	Readily exclusive - "it is written… SERC fridge rules";	Achievable by workaround - "you can just write anyone's name from SERC"	A workload assigned to them: Saturday disposal rule made by management, not them
Coffee machine	A privilege lost by student failure to maintain it	Seen simply as "not for students" - eligibility assumed, not questioned	Always professor-only since they arrived - history unknown to them	Duty and enforcement burden - staff refill and the guard turns students away

GPUs	A peer-negotiated queue - "request other try to find the workload within themselves"; website prevents blind kills	Mostly invisible; UG students "at a little distance disadvantage"	A Slack persuasion market - "convince someone that I want it more, and if they agree"	-
Raising problems	"First talk to the person… it does go to professors" when unsolvable	"Why would I bother anyone about a very small situation?"	Solve among ourselves; only unresolved cases reach PK	Have their own clear chain - tell admin, inform supervisor - that students lack

Alternative Problem Framings

Framing	Design space it opens/ Design Opportunity	Limitation
"Capacity is scarce - the commons needs more desks, GPUs and amenities"	Procurement, space expansion, added capacity	Contradicted by the data - empty-claimed desks and GPUs "held in advance for someday" show availability failing, not supply
"Resources and demand never see each other - empty desks, finished runs, and free slots remain invisible to seekers, and unmet need remains invisible to authority"	Status/booking systems, expiry timers, release rules, complaint-routing, periodic surveys, an occupancy register	Doesn't account for juniors who see an available seat and still won't sit in it
"Access is governed by personal permission and social confidence - those with connected networks work; those without yield silently"	Structured borrowing protocols, junior voice channel, named access rights that don't depend on knowing the holder	Full formalization would break the courtesy network the system currently survives on; some permissioning is genuinely valuable (protecting equipment, setups)

Reframed Problem Statement

Shared resources in SERC are not only constrained by limited capacity; they are made harder to access by fragmented ownership, fragmented visibility, unwritten rules, and dependence on informal negotiation. This means a resource can be physically available but still remain effectively unavailable because users may not know its status, may not know whether they are allowed to use it, or may hesitate to ask. The result is avoidable waiting, under-utilization, unequal ease of access, and repeated coordination effort.

Evidences:

- **Resource ownership is fragmented:** compute and other resources are often controlled by individual professors or groups, so each group develops its own access process.
- **Information is fragmented:** some groups use Slack, some use trackers, some rely on direct communication, so there is no single view of availability or access.
- **Rules are often informal:** many expectations are communicated through word of mouth, senior knowledge, or local norms rather than a common documented system.
- **Access depends partly on social coordination:** juniors often approach seniors first, and desk/resource use may require asking the usual user or negotiating with whoever currently holds it.
- **Physical availability is not the same as usable availability:** desks can be empty but socially claimed, and compute can remain occupied even when not actively used.